

USER CLUB

Digital twins are in a nascent stage in India. Here are some early adopters



► **Piramal Glass** uses Microsoft's Azure IoT to optimise its manufacturing operations and create a feedback loop between its quality control and production teams



► **Cairn Oil and Gas** in India has deployed Honeywell Forge enterprise performance management (EPM) software for improving efficiency, while enabling workers to remotely operate the facility



► **Nokia** is leveraging digital twin technology to quickly monitor and assess the impact of 5G implementation, while providing automated recommendations



► **VE Commercial Vehicles** has adopted Dassault Systèmes 3DEXPERIENCE platform to cost-effectively develop and deliver innovative, high-quality trucks and buses



“Due to the push for digitisation, Asia Pacific is likely to capture a considerable share [of the digital twin market]”

PRAVEEN MYSORE

DIRECTOR-INDUSTRY PROCESS CONSULTANTS AND SOLUTIONS, DASSAULT SYSTÈMES INDIA



“Carbon digital twins can help in visualising the reduction of the carbon footprint of equipment and processes by analysing data”

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like IIoT, connectivity (4G, 5G, 6G), cloud computing, AI & ML and sensorisation of asset/entity, but digital twin is a necessary foundation over which future ‘as-a-service’ business models will be built.”

Typical PoCs or one-plant pilots cost \$200,000-500,000, depending on the size or scale of operations and the complexity of the model. Plus, there is the cost of maintaining the digital twin along with the necessary IT infrastructure. The payback period is typically 6-12 months for PoCs with the RoI in certain cases being more than 10x, says a Wipro spokesperson. Take the example of Piramal Glass, which uses Microsoft's Azure IoT to optimise its manufacturing operations and create a feedback loop between its quality control and production teams. The company is tracking every part of the process from the time the raw material is fed into the furnace to the time its bottles come off conveyor belts. “With Azure Digital Twins, Piramal Glass has achieved 40 per cent reduction in manual data gathering, 25 per cent improvement in employee productivity, and 5 per

cent reduction in defects,” says Shivir Chordia, Azure Business Group Lead at Microsoft India. “The RoI is OPEX reduction, improved productivity due to automation, and better run-and-operate models,” explains Nikhil Malhotra, Tech Mahindra's Global Head of Makers Lab. The IT company had created a complete training landscape for its energy customers in India where a digital twin not only is a digital representation of the landscape but also enables training of company associates around plant operations, enabling OPEX reduction and improved productivity of associates.

For companies, adopting this technology has several advantages like reduced OPEX, improved productivity due to automation, and even discovering errors that are otherwise difficult to find; but the technology can be leveraged to its full potential only when challenges such as availability of quality and real-time data, reliable connectivity and infrastructure, and upskilling and reskilling are addressed. **BT**

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